Urban Heat Mitigation Decision-Support Toolkit for Omaha

Project Mockup and User Testing

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1 Scenario of Envisioned Use

Sarah, a City of Omaha planning analyst, has been asked to identify areas where heat mitigation investments will provide the greatest benefit to vulnerable populations. She opens the Urban Heat Mitigation Toolkit to explore the city's heat and social vulnerability data, understand why certain areas are prioritized, and compare how the results change under different planning scenarios.

Using the map and side panel, Sarah selects a high-priority area in North Omaha. The toolkit shows the priority score, the contributing criteria, and suggests potential interventions. She then adjusts the scenario weights to emphasize canopy deficits and planting opportunities, allowing her to see how the priority areas and recommendations change. With this information, Sarah can prepare a more targeted and equitable heat mitigation plan for the city.

2 Project Mockup (High-Fidelity Wireframe)

The mockup above shows the key elements of the proposed application:

1. Interactive map of Omaha with priority areas, heat layer, and basemap controls.
2. Layer selector to toggle between relevant datasets (heat, canopy, impervious surfaces, population, etc.).
3. Priority area panel showing the overall score, criteria contributions, and recommended interventions.
4. Scenario controls to adjust planning focus and weights.
5. Quick insights with summary statistics for the selected area or view.

The layout is conceptual and focuses on how information will be communicated to the user. It is not a fully functional application.

3 User Testing Plan

The goal of the user testing is to determine whether the mockup is easy to use, provides the right information, and supports informed decision-making.

Test Tasks

1. Locate a high-priority area in North Omaha.
2. View the criteria contributions for the selected area.
3. Change the scenario to emphasize canopy and planting opportunities and note how the priority area changes.
4. Identify one recommended intervention for the area.

Sample Questions

- Was it easy to find a high-priority area?
- Do the criteria contributions help you understand why the area is prioritized?
- Does the scenario control make sense and feel intuitive?
- What additional information would help you make a decision?

4 Mockup Summary and Reviewer Instructions

What this mockup represents

This mockup shows the intended look and feel of the Omaha Urban Heat Mitigation Decision-Support Toolkit. It demonstrates how a planner will explore the map, review priority information, and compare scenarios.

What is conceptual or simulated

- The map data, priority scores, and statistics are illustrative (not real results).
- The application is a static wireframe and not a working prototype.
- The specific datasets, analysis methods, and visual styles may evolve.

How to review

Please explore the mockup as if you were an Omaha planner. Consider the following and provide feedback:

- Is the information clear and easy to understand?
- Does the layout support your workflow?
- What would you change or add?
- Are there any features or information you're missing?

This mockup represents a first pass at the overall vision for the Urban Heat Mitigation Decision-Support Toolkit for Omaha. The layout, interface, features, and presentation of information are conceptual and subject to change as I receive feedback and move further into the project's data processing, analysis, and application development. At this stage, the goal is to communicate the general direction of the project and how users might interact with the tool to explore urban heat, understand potential priority areas, and compare different mitigation scenarios. Feedback on the layout, usability, and information presented will help guide future revisions as the project develops.

Scenario of Envisioned Use: Omaha Urban Heat Mitigation Decision-Support Toolkit
Persona: Sarah, a municipal planning analyst working with the City of Omaha's sustainability and urban forestry team.

Planning context: Sarah has been asked to help identify areas where urban heat mitigation investments could provide meaningful benefits. The city needs to consider not only where surface temperatures are high, but also where tree canopy is limited, impervious surfaces are prevalent, and populations may be exposed to extreme heat.

Sarah opens the Urban Heat Mitigation Decision-Support Toolkit to explore the geographic distribution of these factors. She begins by examining the heat layer and then enables tree canopy, impervious surface, and population exposure layers to understand how environmental conditions and potential human exposure overlap.

After identifying an area that appears to be a priority, Sarah selects it on the map. The Priority Area Details panel displays its overall priority score and explains which criteria contribute to the result. This helps her distinguish between areas prioritized primarily because of high heat intensity and areas where heat, population exposure, and canopy deficits combine.

Sarah then compares two planning scenarios. The first emphasizes heat intensity and population exposure, while the second emphasizes canopy deficits and planting opportunities. She observes whether the same areas remain priorities or whether the emphasis on different criteria changes the geographic distribution of results.

Using this information, Sarah identifies areas that warrant further investigation and considers which mitigation approaches, such as tree canopy expansion, shade structures, or cool surfaces, may be appropriate for additional evaluation. She also reviews the underlying criteria and data limitations before communicating the findings to colleagues.

The toolkit helps Sarah explain why certain areas receive higher priority, understand how planning objectives influence the results, and support a more transparent discussion of potential investments. The results inform planning decisions rather than replace detailed site assessments, community engagement, or professional judgment.

Project Mockup Summary and Reviewer Instructions

Project Overview

The Urban Heat Mitigation Decision-Support Toolkit for Omaha is intended to help municipal planners explore urban heat conditions and identify areas that may warrant further consideration for heat mitigation investments. The proposed application integrates environmental, demographic, and geographic information to explain priority areas and compare how results change under different planning objectives.

What the Mockup Represents

The wireframe illustrates the proposed interface and how its main components will communicate spatial information. It presents a map-centered layout with environmental data layers, a priority-area details panel, configurable planning scenarios, and summary information.

The design emphasizes three primary user tasks: exploring the distribution of urban heat and related conditions, understanding the criteria contributing to an area's priority, and comparing results under different planning scenarios. Potential intervention categories are included to connect analytical results to practical planning discussions.

How to Review the Mockup

Please review the wireframe as if you were an Omaha planner using the toolkit to investigate where heat mitigation investments might be most beneficial. Follow the user-testing tasks provided and consider whether the proposed layout supports each task.

Focus on the clarity of the map, the organization of the information panels, the explanation of priority scores, and the usefulness of the scenario controls. Consider whether the design makes it easy to understand why an area is prioritized and how changing planning objectives might affect the results.

The wireframe is conceptual. Some controls and interactions are proposed rather than operational, and displayed scores or statistics are illustrative rather than actual analytical results.

Please provide feedback on what is clear, what is confusing, what information appears to be missing, and what changes would make the toolkit more useful for planning decisions.

User-Testing Plan
Objective

Evaluate whether the proposed interface allows users to explore urban heat conditions, identify potential priority areas, understand the factors contributing to their prioritization, and interpret differences between planning scenarios.

Instructions for Testers
Assume you are an Omaha municipal planner investigating potential locations for future heat mitigation investments. Use the mockup to work through the tasks below. Some elements are illustrative or simulated, so evaluate the intended layout and interaction rather than expecting functional controls.

Please complete the tasks in order and answer the questions based on your experience. If you cannot determine how to complete a task from the mockup, indicate where you became uncertain rather than guessing.

Tasks and Questions
Task 1: Explore heat conditions
Examine the map and layer controls. Identify an area that appears to have high heat exposure or other conditions that may warrant further investigation.

- Was it clear which map layer represented heat intensity?
- Could you distinguish the priority areas from the underlying environmental data?
- Was the legend or symbology easy to interpret?
- What additional geographic context would help you evaluate the area?

Task 2: Understand a priority area
Select or imagine selecting a priority area and examine the proposed details panel.

- Could you determine why the area received a high priority score?
- Were the individual criteria and their contributions understandable?
- Would you want to see additional information about the population or environmental conditions in that area?
- What information would you need before discussing this area as a potential investment priority?

Task 3: Compare planning scenarios
Examine the proposed scenario controls. Compare a scenario emphasizing heat intensity and population exposure with one emphasizing canopy deficits and planting opportunities.

- Was it clear what each scenario was intended to prioritize?
- Could you understand how changing the planning focus might change the results?
- What additional visual comparison would help you identify areas that changed in priority?
- Would you prefer preset scenarios, adjustable weights, or both? Why?

Task 4: Interpret potential interventions
Review the proposed intervention categories associated with a priority area.

- Did the interface communicate the relationship between the analysis and potential mitigation approaches?
- Was it clear that the suggested interventions require further evaluation rather than representing definitive recommendations?
- What additional site information or constraints would you want to see?

Overall Feedback

1. What part of the proposed interface was easiest to understand or use?
2. What part was most confusing or required the most interpretation?
3. Was any important information missing from the map or details panels?
4. Did the interface help you understand both why an area was prioritized and how priorities might change under different planning objectives?
5. What is the single most important change you would recommend before development continues?

Evaluation Approach

Feedback will be reviewed for recurring usability issues, unclear terminology, missing information, and opportunities to improve the visual hierarchy. Findings will be used to refine the interface and prioritize features for implementation. Because this is a conceptual mockup, the evaluation will focus on the proposed workflow and communication of information rather than analytical accuracy or application performance.

Citations

Omaha-Council-Bluffs Metropolitan Area Planning Agency. (n.d.). *Omaha-Council-Bluffs Metropolitan Area Planning Agency*. https://www.omacap.org/

GitHub. (2026). *GitHub Copilot* [AI-powered coding assistant]. https://github.com/features/copilot